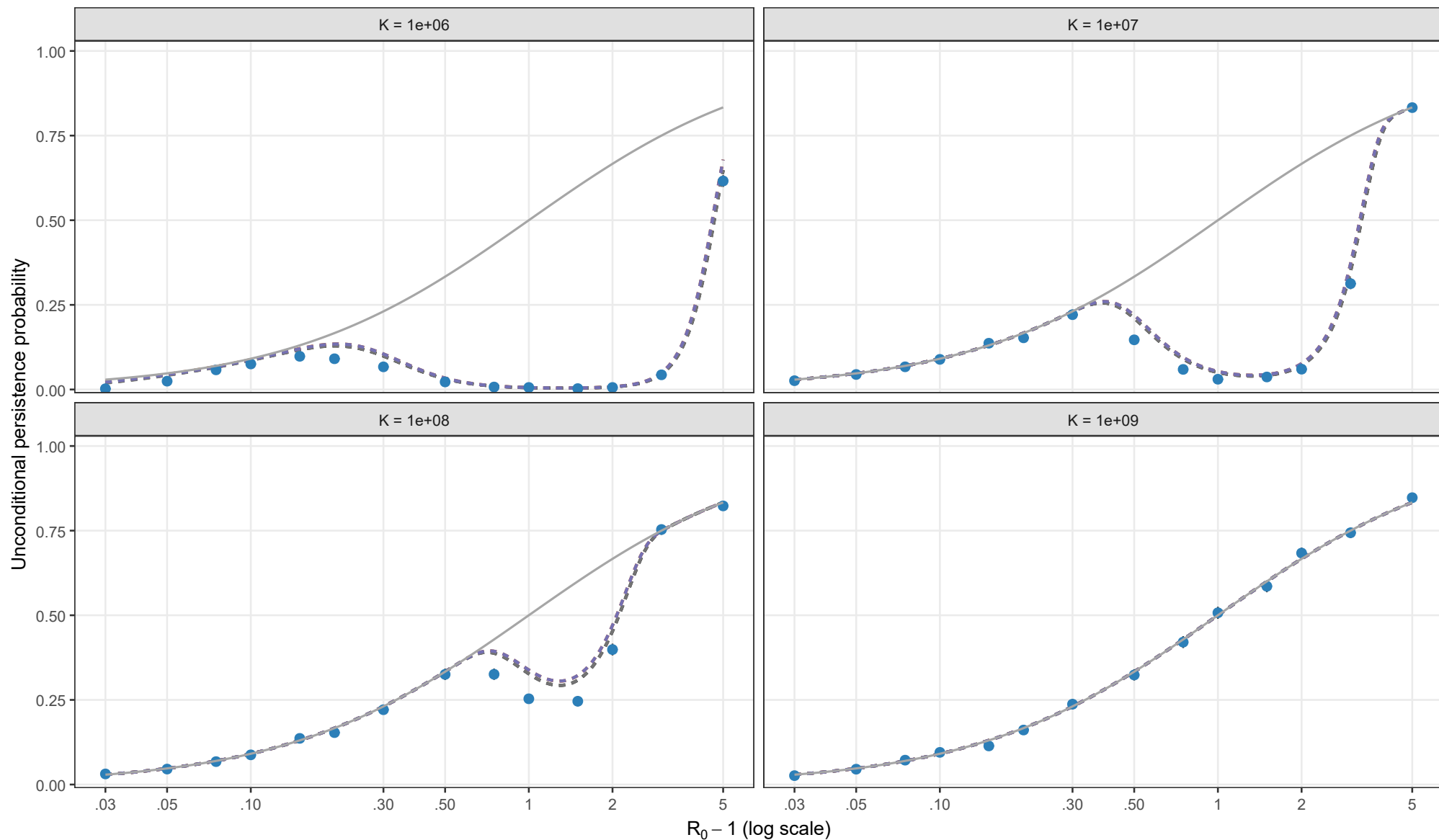


Unconditional persistence probability

$\rho = 0.01$, $\theta = 0$, $I_0 = 1$; matching height = y^*



Analytical method

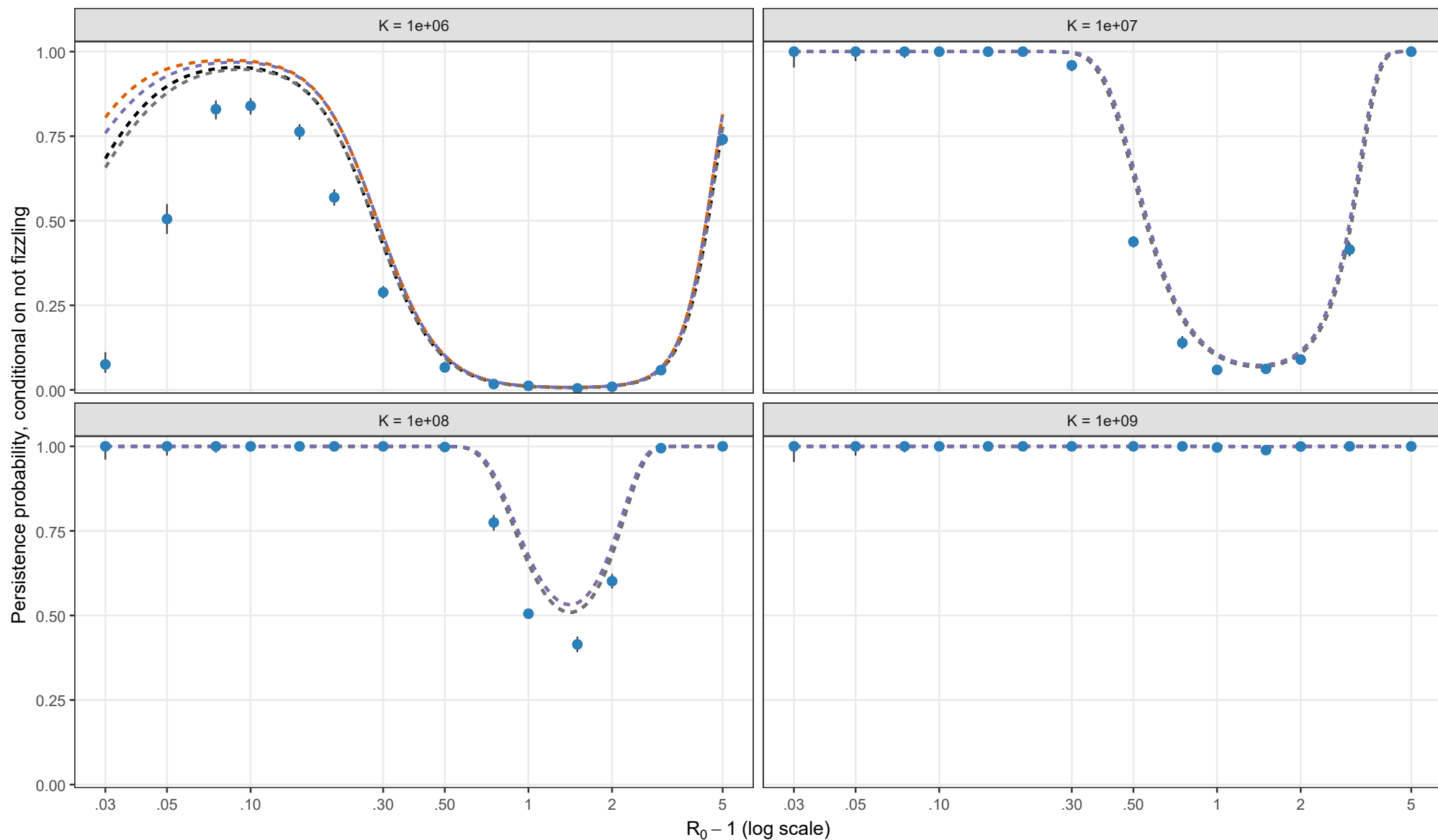
- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry

Matching height $y_{BL} = y_{star}$

Points and 95% Wilson intervals: adaptive tau-leaping (epsilon = 0.01); 3,000 attempts per point, expanded to 10,000 when needed. All analytical curves use matching height y^* . Curves use the analytical D_θ expression. Missing curve segments indicate no admissible matching root.

Persistence probability conditional on escaping early fizzle

$\rho = 0.01$, $\theta = 0$, $l_0 = 1$; matching height = y^*



Analytical method

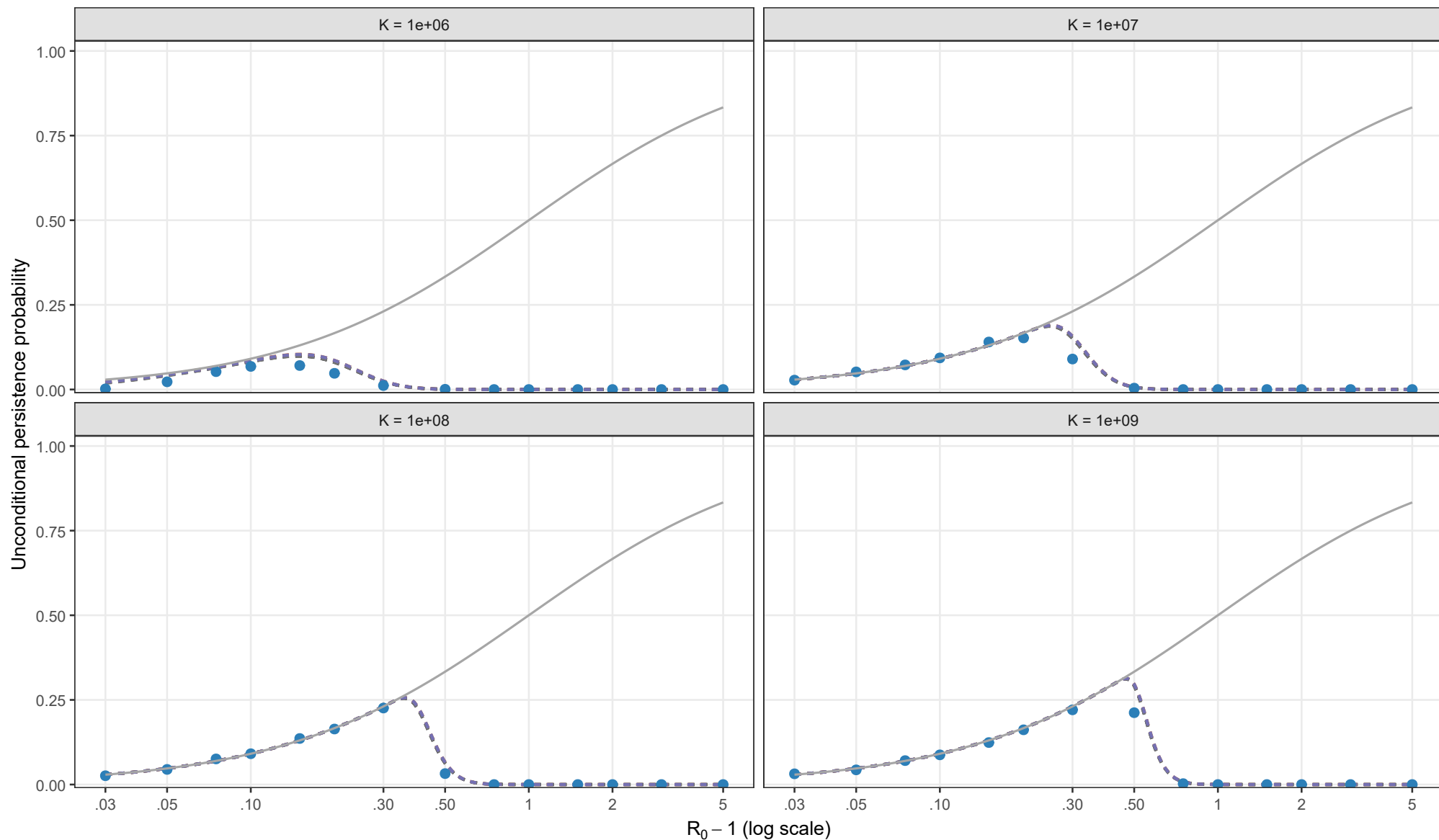
- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry

Matching height $y_{BL} = y_{star}$

Points and 95% Wilson intervals: adaptive tau-leaping (epsilon = 0.01); 3,000 attempts per point, expanded to 10,000 when needed. All analytical curves use matching height y^* . Curves use the analytical D_{θ} expression. Missing curve segments indicate no admissible matching root.

Unconditional persistence probability

$\rho = 0.01$, $\theta = 0.5$, $l_0 = 1$; matching height = y^*



Analytical method

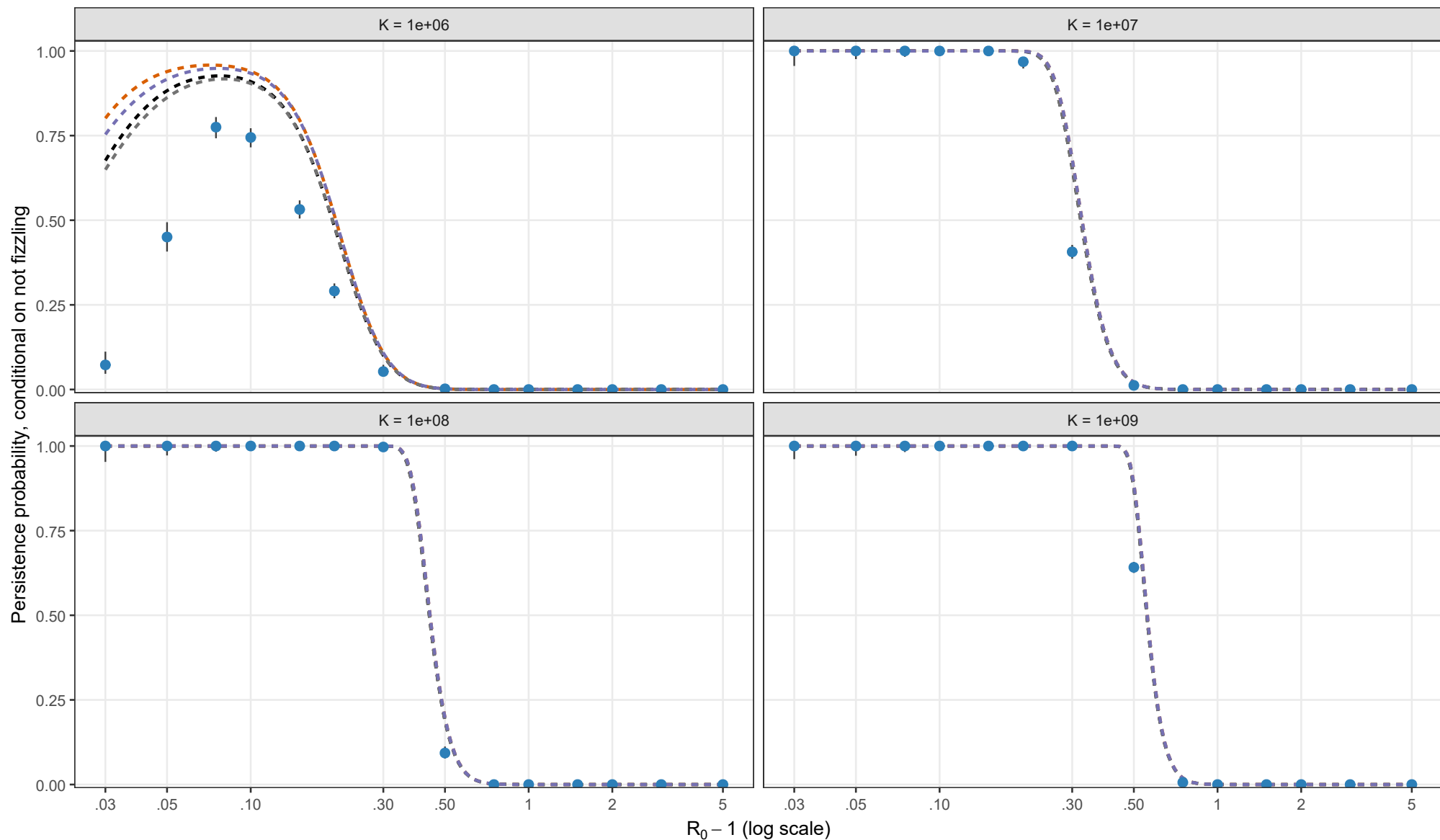
- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry

Matching height $y_{BL} = y_{star}$

Points and 95% Wilson intervals: adaptive tau-leaping (epsilon = 0.01); 3,000 attempts per point, expanded to 10,000 when needed. All analytical curves use matching height y^* . Curves use the analytical D_θ expression. Missing curve segments indicate no admissible matching root.

Persistence probability conditional on escaping early fizzle

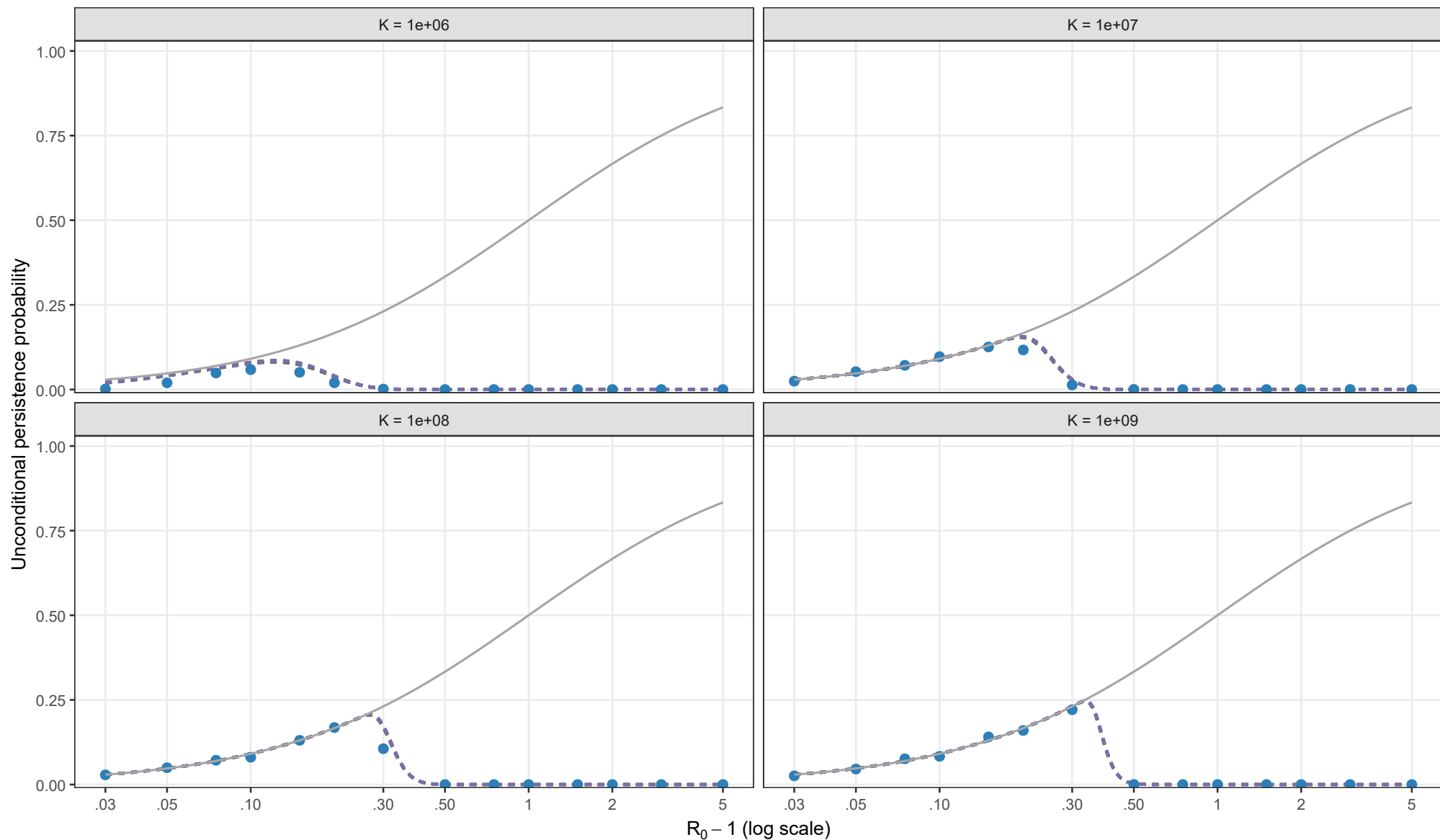
$\rho = 0.01$, $\theta = 0.5$, $l_0 = 1$; matching height = y^*



Points and 95% Wilson intervals: adaptive tau-leaping (epsilon = 0.01); 3,000 attempts per point, expanded to 10,000 when needed. All analytical curves use matching height y^* . Curves use the analytical D_θ expression. Missing curve segments indicate no admissible matching root.

Unconditional persistence probability

$\rho = 0.01$, $\theta = 1$, $I_0 = 1$; matching height = y^*



Analytical method

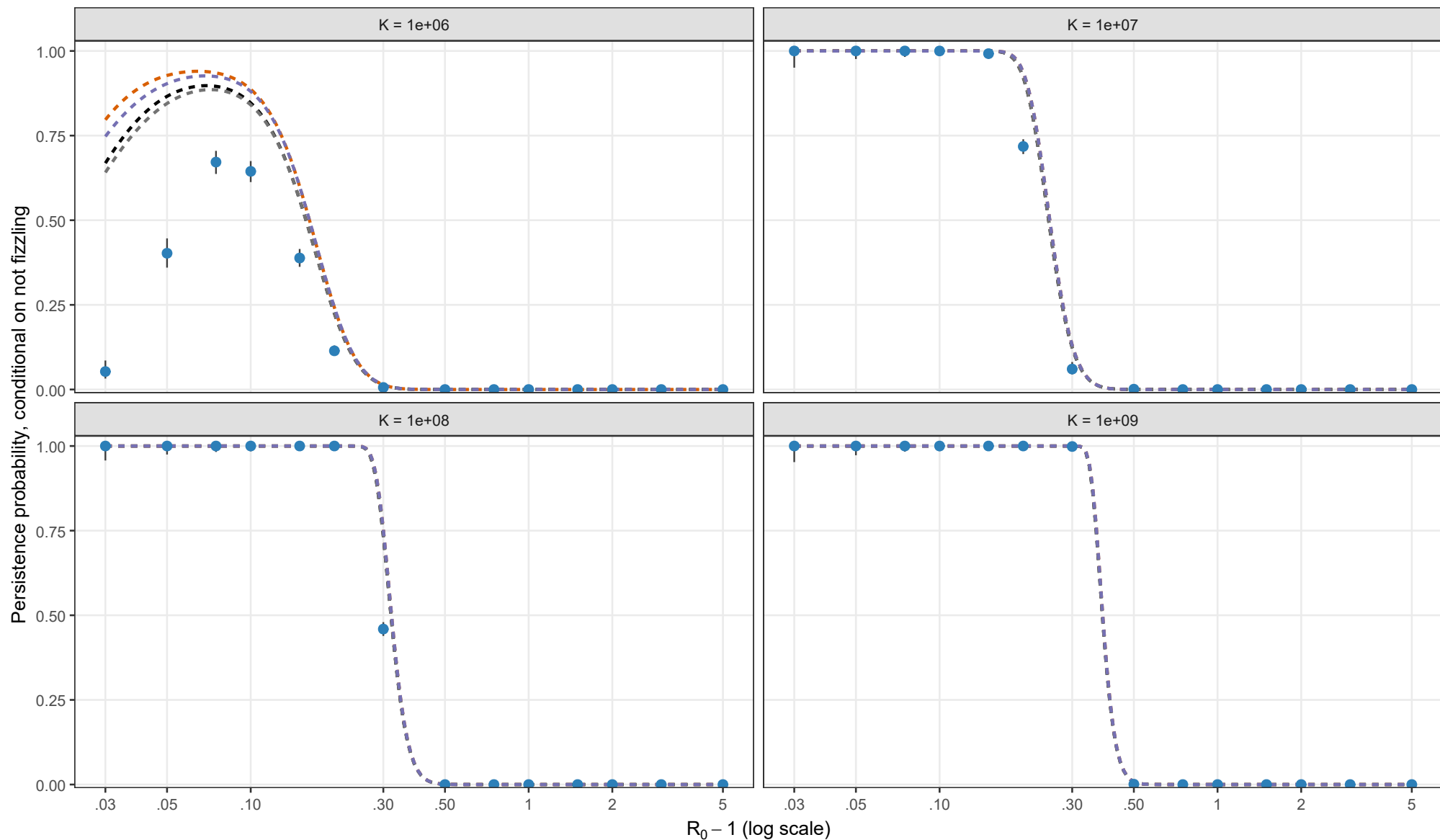
- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry

Matching height $y_{BL} = y_{star}$

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